

2LA Module information 2023

This information sheet should be read in conjunction with the MAM2000W+ Course Information Sheet.

- **Lecturers:** Mr Simon Chili and Dr Holger Spakowski (module leader)
 - **Offices:** Mathematics Building, room M3.00 (SC) and room M1.02 (HS)
 - **Consultation hours:** Consultation times will be announced shortly.
 - **Email address:** `simon.chili@uct.ac.za` and `holger.spakowski@uct.ac.za`
- **Syllabus:** We will first look at matrices, explaining how to use them to solve systems of linear equations and studying their properties (inverses, determinants, and so on). Then we will define vector spaces and examine some of their properties (subspaces, linear independence, bases). After that, we will define linear maps between vector spaces; these are the maps that preserve some algebraic structure of vector spaces. Particular topics in this area include the image space and kernel of a linear mapping, the rank-nullity theorem, and the effect of changing bases. We will then study the matrix representation of linear transformations (column space, row space, change of basis, and similar matrices). A highlight of the course is the study of determinants, eigenvalues, eigenvectors, and diagonalization of matrices. These concepts have numerous applications in applied mathematics. In the end, we will cover inner product spaces and delve into the topic of Gram-Schmidt orthogonalization.
- **Lectures:** Lectures will be Mondays, Thursdays, and some Wednesdays at noon: HS in Maths 320 and SC in Maths 304. The Wednesday lectures will be on the following days: 15 Feb, 1 Mar, 15 Mar, 12 Apr, 26 Apr, 3 May, and 10 May.
- **Notes:** You will find a detailed set of notes for the module in Linear Algebra Resources on Vula (`2LANotes23.pdf` under “Notes”). We will be following these notes quite closely in lectures. The exercises in the typed notes are essential to deepen and check your understanding of the material. Therefore, you should do the exercises as you read, and we will post the solutions at the end of each chapter.
- **Tutorials:** A tutorial will be posted on Vula every week. This is homework, but you will have the opportunity to get help and ask questions in a weekly tutorial on either Thursday or Friday afternoon. There are two one-hour long slots on each day (14h00–15h00 or 15h00–16h00), and you can sign up for a slot on Vula. Please attend your assigned tutorial so that none of the sessions become overcrowded. (We will not provide printed tutorial sheets for 2LA at the tutorial venues.) The solutions to the tutorial will be posted on Vula on Friday afternoons, and you should go through them carefully. **It is crucial that you stay up to date with the tutorials:** they are designed to complement the notes and deepen your understanding of what you are doing in class that week, so you cannot get the same effect by doing the tutorial later. Tutorials are compulsory, and we will take a register. You need to attend at least 75% of the tutorials to get a DP.
- **Homework Tests (Quizzes):** To ensure that you stay current in this module, you are required to complete weekly homework tests, which will contribute 15% towards your 2LA class record. We will be using software developed by the Mathematical Association of America. The name of the program is WeBWoRK; it is similar to, but not the same as, WebAssign, with whom many of you may be familiar.

The assignment for each week will be based on the questions in the 2LA tutorial for that week. The assignment will become available on Tuesday afternoon at 17h00, and you will have time until 23h59 on Wednesday to complete the assignment. You will be allowed three attempts at each assignment (but the questions will be different each time), and we will use the best of your two marks for the WeBWorK part of your class record.

There are detailed instructions for using WeBWorK on Vula. Please read the instructions **How to use WeBWorK for 2LA** that is available under the Linear Algebra Resources on the MAM2000W Vula website. Please do not attempt to log in to the site before we make an announcement that it is ready.

- **Assessment:** The linear algebra module is part of MAM2000, and you will write a final exam in June (exact details to be announced). You will write two class tests, which will be in the evening from 18h00 to 19h30, each of which is worth 42.5% of your class record. The weekly homework assignments account for the remaining 15% of your 2LA class record. The final exam will be two hours long.
- **Class tests:**
 - **Test 1:** Monday, 13 March 2023, 18h00–19h30
 - **Test 2:** Wednesday, 3 May 2023, 18h00–19h30

Please check these dates carefully to make sure that you do not have a clash with another scheduled test. If you do have such a clash, you must send an email as soon as possible, but not later than Friday, 17 February, to Dr Spakowski (holger.spakowski@uct.ac.za), giving the name of the course which has a test at the same time and the exact time and duration of this test. We need an email from **every student affected** so that we can know the number of students involved.

If you miss a test for good reason, please send an email to holger.spakowski@uct.ac.za with an explanation (illness, death in the family, mugging, etc.) along with some evidence (medical certificate, death certificate, case number, etc.) as soon as possible, but not later than a week after the test. That email should be copied to the MAM2000W course convenor (Thomas van Heerden) as well.

Please note that departmental policy is to accept medical certificates that state “According to my knowledge” and not to accept certificates that state “As I was informed”. Therefore, please ask your practitioner to be clear on the certificate.

If you miss one test for a valid reason, you will be excused from that test, and your mark for the remaining test will be doubled when calculating your module record. If you miss both tests, you will have to sit an alternative assessment before the exam. This is normally an oral test, and your module leader will liaise with you to arrange a mutually suitable time.

- **Final mark:** Your final mark for each module will be either 40% class record, 60% exam mark, or 20% class record, 80% exam mark (whichever is better for you). Your final mark for the course is the average of your final marks for each module.
- **Words of advice:** This course in linear algebra is likely the first time you will need to think about mathematics abstractly and the first time you will be asked to (carefully) prove mathematical statements. Also, lectures in this module will proceed at a markedly faster pace than the MAM1000 lectures, posing an additional challenge. Our aim is for all of you to achieve your full potential in this new year, so we will offer some guidance to help set you on the right path.

Most importantly, you must remember that **mathematics is not a spectator sport**. To succeed in this course, you must attend lectures and tutorials and work through the tutorial problems every week. You should read ahead of the lectures. We will cover the material in the lecture notes in that order, so you should never be completely surprised by the lecture material. Next, you need to work through as many problems as possible, including the tutorial problems, and begin working on these problems as soon as possible. It might help to form study groups and work through problems together.